Dhyan Vasudeva Gowda

Software Engineer | Healthcare AI and Connected Medical Devices | LLM Systems on AWS
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Experience

Computer and Information Research Scientist (Software Engineering and Applied AI) Feb 2026 – Present
MG Health Tech Inc. *McKinney, TX*

- Developing the capture engine for a home sleep testing mobile app, recording overnight SpO2, pulse rate, a 125 Hz PPG waveform, and activity from a Bluetooth pulse oximeter, with fault-tolerant background capture and session recovery
- Built the sleep-study server and data processing pipeline (Python, AWS Lambda) that verifies uploads by SHA-256, stitches 15-minute segments, checks quality, and converts packet logs into European Data Format (EDF) for EnsoData sleep scoring
- Designed the API linking the sleep-study server to the MySleepReport physician portal (mysleepreport.com): study orders, EDF handoff through short-lived download links, and scoring outcomes
- Collaborated on integrating Bluetooth Low Energy (BLE) medical devices (blood pressure monitor, ECG recorder) into the VitalWeave React Native app (vitalweave.ai; App Store, Google Play), sending readings to physicians
- Co-built a real-time AI voice agent (asyncio, Deepgram ASR, GPT-4.1-mini function calling, Telnyx) that answers calls 24/7 for multiple clinics, handling appointment requests, refills, cancellations, payment inquiries, and clinic questions
- Engineered the voice agent's HIPAA and TCPA safeguards: two-phase SMS consent, append-only audit logs, protected health information (PHI) masking, signed webhooks, and fail-closed opt-outs
- Productionized the AroCord electronic health record (EHR) summarizer: GPT-4o summaries checked by GPT-4o-mini, pgvector similarity search, Amazon Comprehend Medical NLP entity extraction, and LLM cost quotas in DynamoDB
- Prototyped a VitalWeave wellness analytics dashboard that turns wearable ECG/PPG signals into daily recovery, strain, sleep, stress, and HRV scores using open-source signal processing and ML, with an AI chatbot grounded in each user's data
- Architected the production multi-tenant SaaS data model in MySQL, with role-based access control (RBAC), audit logging, and billing, serving the VitalWeave patient app, the VitalWeave clinician platform, and MySleepReport
- Provisioned the distributed systems backend on AWS for VitalWeave, MySleepReport, and AroCord: serverless REST APIs (Lambda, API Gateway), the voice agent on EC2, SQS/SNS, RDS, S3, Cognito, KMS, Secrets Manager, and CloudWatch
- Authored the technical plan for hospital private 5G edge AI and 6G contactless sensing with a global telecom vendor

Software Development Intern Jun 2025 – Aug 2025
MG Health Tech Inc. *Remote*

- Built a patent-pending AI system (Python, FastAPI, React, GPT-4o) that summarizes patients' medical documents in under a minute, cutting clinic intake by up to 15 minutes, with PDF, QR code, mobile wallet, and FHIR-ready outputs
- Implemented a safety guardrail that evaluates each summary against source documents and removes hallucinated claims
- Deployed the multimodal MedGemma model for medical image analysis behind a private Hugging Face inference API
- Created a source-verified RAG assistant (LangChain, FAISS, Llama 3) that answers doctors' questions from patient records

Teaching Assistant – Generative AI and Software Project Management (two semesters) Jan 2025 – Dec 2025
Illinois Institute of Technology *Chicago, IL*

- Guided more than 100 graduate students building generative AI applications; developed AutoGen, LangGraph, and LlamaIndex labs and demoed React/Flask microservices, Google Cloud deployments, and TensorFlow LSTM models
- Mentored students on RAG, vector embeddings, Elasticsearch, Docker, agentic AI, and LLM API integrations

Software Development Intern Feb 2023 – Nov 2023
Torus Solutions *Mysuru, India*

- Built a full-stack food donation platform (Java, React, MySQL) with multi-role authentication and NGO–donor matching

Patents (Co-inventor, Pending)

System and Method for Medical Document Processing | India 202521133823 A; US Appl. 19/653,187
System and Method for Closed-Loop Patient Monitoring and Predictive Clinical Decision | India 202641079234 A
System and Method for Deterministic, Contactless Patient Monitoring and Predictive Clinical Decision Support | India 202641079235 A

Education

Illinois Institute of Technology, Chicago, IL | *Master of Science in Computer Science, GPA 3.91/4.0* | Dec 2025
JSS Science and Technology University (SJCE), Mysuru, India | *B.E. Computer Science, CGPA 8.66/10* | Jul 2023

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, HTML, CSS
AI/ML: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), prompt engineering, NLP, vector databases (pgvector, Pinecone, FAISS), agent frameworks (LangChain, LangGraph, AutoGen, LlamaIndex), Hugging Face, TensorFlow
Cloud: AWS, serverless, Lambda, API Gateway, EC2, RDS, S3, Cognito, SQS, SNS, KMS, CloudWatch (observability), GCP
Databases: MySQL, PostgreSQL, NoSQL (DynamoDB, MongoDB), Elasticsearch
Frameworks and Tools: React, React Native, Node.js, FastAPI, Flask, REST APIs, pytest, Docker, Git, BLE, Twilio, Telnyx
Healthcare: HIPAA, protected health information (PHI), BAAs, HL7, FHIR R4, ICD-10, remote patient monitoring (RPM)